

RESPONSE TO THE REFEREE ON NEURAL BELLMAN OPERATORS

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The author thanks the referee for separating the accuracy of a returned action, the construction of its neural continuation, and the complete work needed to obtain it. The present revision retains Neural Bellman Operators, the original controlled-economy problem, and all economic applications. It develops the neural construction and its full-policy implication rather than replacing the paper with a method-neutral certification paper. The original article, technical supplement, adverse numerical outcomes, and applications are preserved as explicitly identified companions.

The report addressed is the latest independent R21 report dated 5 October 2026, on the review branch ending in 2789166. The immutable review commit, report blob, reviewed R21 commit, inherited derivation commit, new science-freeze commit, and new execution identity are recorded in the release audit. The report is an owner-commissioned advisory exercise, not an editorial decision issued by Econometrica. The response below distinguishes results already derived after R21 from the new R37 precision theorem and executed comparison.

PRINCIPAL CHANGES AND READING ORDER

The current article now contains the original model and NBO method, decision-centered errors, full-policy comparisons, primitive-budget hidden-weight construction, the mixed-norm warm-start policy bound, and the new precision-adaptive refresh theorem. The active technical supplement supplies the corresponding proofs, together with the general composition argument, gradient-curvature criterion, and noncommuting linear warm-start construction. The previous article and supplement remain complete historical companions; the applications remain complete and separately accessible. The publication is organized by the economic and mathematical argument, not by revision numbers.

The new theorem gives a whole-step precision allocation under which the relative Gram-error enclosure squares at every update. It permits rectangular hidden factors and noncommuting initial matrices. It supplies a finite update cap and a sum-of-fractional-precisions bound, while allowing a fixed target inverse to be constructed once within each fit. Its full-policy corollary pays for fitting, own-policy evaluation, and actor error separately. The underlying ideal polar iteration is classical and credited to the matrix-functions literature; the economic precision allocation and its connection to the implemented policy are stated and proved explicitly.

The new experiment was source-frozen before execution. Four procedures each construct an anchor policy and a changed-valuation policy on the unchanged inherited exact construction primitives. All eight services pass the same full-policy tolerance of one ten-thousandth. The adaptive-cached neural construction reduces the fractional-precision sums from 240 to 102 and from 120 to 54 relative to fixed-precision quadratic refresh. The recorded complete service clocks are also lower in both specified regimes. The structural solution remains faster than all neural constructions. All outcomes are retained, and neither these eight objects nor algebraic tests are counted as population observations.

RESPONSES TO BLOCKING CONCERNS

B1. The original future-change experiment does not establish neural work superiority

The original result is retained exactly: the old check-and-refresh procedure costs more than fresh neural refitting in all eight dimension-volume cells, and cached simulation is cheapest among the fully certified procedures. We do not reinterpret the old clocks or replace the failed reuse checks.

The affirmative response has two parts. The construction theorem now specifies when an old neural factor is admissible for a changed continuation and how much precision and learning suffice for a full-policy allowance. The new frozen comparison then measures an implementation of that rule against both inherited neural procedures and a strong structural procedure. The adaptive implementation is faster than the two inherited neural procedures in both specified services, while the structural solution is faster still. This is a demonstrated improvement in the implemented neural construction, not a claim of universal neural cost dominance.

A broad frontier over richer economies and candidate families remains an empirical question; the paper does not declare it settled by these two cases.

B2. Certified signs do not establish neural necessity

We retain the finite-law signs with their original interpretation. They establish that the returned actions are accurate enough to sign the stated current-withdrawal comparisons. The same target is available through enumeration, more sharply and cheaply in those experiments. The current article states this explicitly.

The strengthened neural result is instead constructive: trainable hidden weights, updated under an explicit schedule, deliver a continuation allowance that enters an all-adapted-policy certificate. The new exact experiment exercises that implication with changing futures and actual reused factors. A certificate for a neural candidate is not presented as proof that only a neural method can solve the economy.

B3. The composition theorem needs an executed full-policy bridge

The main article now integrates the full-policy and recursive-risk derivations rather than leaving the general composition lemma as an uninstantiated promise. In the recursive capital economy, the local residual is evaluated against the returned policy's own finalized future. The Bellman matrix inequalities cover all real states and all real vector controls. The comparator is all adapted policies of finite recursive cost.

The six accounts requested by the referee are explicit. Own-policy local advantages come from full-action actor residuals. State coverage follows from coefficient inequalities, not a catalogue. Moment margins generate separate continuation weights for quadratic and state-independent errors. Policy-class approximation is zero because the comparison is not restricted to the fitted class. Transfer is zero for the directly specified discrete economy, not for an unverified diffusion approximation. Evaluation, training, actor, and action-execution discrepancies receive their own allowances.

The new exact services independently check a Bellman matrix subsolution as well as the local training and actor conditions. The policy guarantee therefore does

not depend merely on accepting a small fitting loss. Exact rational feedback and its specified arithmetic are distinguished from an arbitrary floating-point implementation.

B4. The original scalar finite-law experiment is too narrow

The original study remains a useful exact laboratory and is not discarded. The integrated full-policy evidence includes 24-date economies, dimensions ten and fifty, continuous Gaussian innovations, and vector controls. Future policies are solved anew in each regime. These features remove finite-support enumeration as a computational procedure and replace sampled-state coverage with an all-state coefficient argument.

The new eight-service experiment has a different purpose: it isolates the whole-step precision and target-cache mechanism in a small exact construction where every check can be reproduced. We do not advertise its two-dimensional instances as a new high-dimensional benchmark. The larger inherited experiments and the small exact construction answer different questions, and both retain the structural comparator.

B5. Fixed-feature regression is not hidden-weight neural training

The constructive results now concern a square-activation scalar continuation with all entries of the hidden factor trainable. The cold construction supplies a primitive envelope, initialization, step size, and finite cap. The gradient–curvature criterion supplies another checked route and excludes the zero-gradient singular saddle under its curvature premise. Linear warm construction permits rectangular factors, noncommuting starts, and bounded update errors.

The new precision theorem strengthens the warm construction by specifying a finite whole-step error schedule and an explicit fractional-precision account. Its proof is not transferred to an unrelated jointly trained architecture. The original smooth-network formulation remains present with its own hypotheses. The matrix polar identity is credited as classical rather than presented as a newly invented iteration.

B6. A small set of seeds does not establish population reliability

The paper now states the deterministic admissible class for the construction theorem. Its implication is uniform over every initial factor in the verified basin and every implemented update satisfying the full error account. This is a mathematical reliability statement, not a confidence interval obtained from repeated fits.

The new comparison has a prespecified finite-object estimand: the exact acceptance outcome, construction counts, precision choices, and complete service clock for eight declared objects. Its sources and protocol precede the observations. Historical seed catalogues retain descriptive counts. Replaying a stored object, checking a matrix inequality, or running an overlapping inherited unit test never increases the number of independent economic observations.

B7. The broad applications require their own decisive conditions

Every original application is retained. The current economic-scope section keeps the distinctions among recursive utility, endogenous preference adjustment, temporal selves, and strategic games. Their utility domains, admissibility conditions, and deviation or equilibrium comparisons are not silently replaced by the Gaussian capital model.

The recursive-risk capital economy supplies one complete constructive full-policy instance, including the placement of discounting outside the certainty equivalent and the Gaussian moment domain. The paper does not claim that this instance numerically establishes neural superiority for each other application. Preserving the breadth of the original paper does not require treating distinct mathematical obligations as already discharged by a common architecture.

B8. The cumulative presentation is difficult to read

The current paper is a genuine new article entry point, not a report pointing to an unchanged R21 root. Its argument runs from the economic problem to decision errors, policy comparison, neural construction, precision-adaptive reuse, and evidence. The active supplement contains the proof chain. The complete old article and

its supplement are labeled historical companions, and the economic applications remain complete.

A preservation map records the original source components and labels. Historical sources are not rewritten or deleted to produce a favorable comparison. Only explicitly identified current copies receive editorial corrections, including the terminal risk transform. The release includes actual native-class PDFs and current source entry points, with compilation and cross-reference diagnostics recorded independently of the scientific claims.

B9. The economic primitives and policy interpretation are stylized

The capital economies are specified theoretical constructions rather than empirical calibrations or estimated reforms. The old signs concern current actions conditional on specified future policies. The current dynamic construction instead finalizes each future policy before evaluating it at the preceding date, and repeats that backward solution in each changed economy.

An old continuation is used only as an initialization when its relation to the new target is certified. It does not substitute for the new policy's own continuation. The terminal actor uses the known transformed coefficient Ψ of the terminal quadratic matrix, not the untransformed terminal matrix except when those coincide. This point is corrected in the current text and retained explicitly in the full-policy argument.

RESPONSES TO REQUESTED CHANGES

M1. Retain one coherent paper

We retain Neural Bellman Operators as the title and subject. The central contribution is a constructive path from economic primitives and trainable hidden weights to full-policy accuracy, including changed futures and explicit arithmetic. Method-neutral certification is used to assess that construction; it is not a replacement topic. The original controlled-diffusion framework and economic applications remain part of the paper.

M2. Instantiate every policy-error term

The full-policy application and the precision-budgeted policy corollary provide the six accounts detailed under B3. In particular, the evaluation radius is spectral and carries the correct dimension factor when combined with Frobenius training error. The separate magnitude gate prevents a factor that overshoots its target from invalidating the primitive policy envelope. The independent exact subsolution check is performed after construction.

M3. Retain hard stochastic and dynamic comparisons

The inherited 24-date, continuous-support, vector-control experiments are integrated with the construction theory. Their analytic Gaussian structure is available equally to the neural and structural procedures. Continuous support is not presented as proof that neural approximation is indispensable. The new exact comparison isolates a learning mechanism and is labeled accordingly rather than replacing the larger design.

M4. Specify the estimand before observing outcomes

The new source-freeze branch contains the materialized science files and the actual protocol, not merely an installer that could later generate different code. Method order, regimes, tolerance, precision caps, failure rule, and complete-cost boundary were fixed before the eight services ran. The evidence branch preserves every detailed record and its hash. No success probability over an unspecified distribution is inferred.

M5. Improve refresh prospectively and retain failed work

The new rule uses a checked relative target-change gate or a direct basin check, then allocates precision as the error bound contracts. A fixed target inverse is owned by one fit and cannot be silently reused for a different continuation. Zero-update acceptance needs no inverse construction. A requested precision beyond the registered cap is an explicit failure, not a certificate. All checks, target applications, and actor work enter the declared cost boundary.

The new comparison is a fresh execution under a prior source freeze. It does not overwrite or retime the original adaptive policy. Failed historical checks remain visible even though all eight new specified services succeed.

M6. Isolate construction and charge complete work

The new table separates hidden updates, training-target inverse constructions, stored fractional precision, full-policy bounds, and complete service time. The underlying records additionally count target-inverse applications, own-policy evaluations, actor solves, independent checks, and simulation transitions. Training-target inverse counts do not pretend to include the inverses needed elsewhere in the policy calculation; those operations remain in the service clock.

Every service clock includes primitive allocation, evaluation, training, checks, actor solves, serialization, and durable output. Common startup and the closing summary are identified separately. Process high-water memory is not relabeled as a per-method memory comparison. Fractional-storage complexity is not advertised as total rational bit complexity. The structural competitor is retained and wins the full-work comparison in the two exact economies.

M7. Separate accuracy, construction, and necessity

The abstract, introduction, theorem discussion, numerical interpretation, and conclusion now use the distinction consistently. A certificate answers what the returned policy achieves. A finite construction theorem answers how the specified trainable continuation reaches its allowance. A complete-work comparison answers which executed procedure is cheaper. The new results strengthen the first two statements and improve measured work relative to inherited neural implementations without asserting neural necessity.

M8. Evaluate the correct future policy

At every date of the new dynamic construction, the future actors have already been finalized and their own continuation is evaluated. Each changed economy has its own backward policy solution. The actual old factors are retained only for

1 initialization. A fixed-target cache is rebuilt for a changed target. The old specified- 1
 2 future comparative statics retain their original interpretation. 2

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 4 *M9. Supply a readable native publication without deleting history* 4

5 The release provides a current Econometrica-class main article, an active technical 5
 6 supplement, a point-by-point response, and complete historical and application 6
 7 companions. Source preservation and cross-document references are audited. The 7
 8 current root entry points refer to the new article rather than R21. Historical build 8
 9 dependencies are taken from the source-bound original publication, not filled with 9
 10 invented tables or observations. 10

11
 12 *M10. Preserve the evidentiary standard* 12

13 The source chronology, immutable service records, full denominator counts, ad- 13
 14 verse comparisons, exact arithmetic checks, and separate execution identity are 14
 15 preserved. The new source suites report 22, 42, and 50 passing test executions; the 15
 16 suites overlap and contain 30 distinct new algebra tests. They are not 114 inde- 16
 17 pendent scientific observations. Compilation, source preservation, mathematical 17
 18 checks, and economic interpretation have separate records. Successful tests and a 18
 19 native PDF are not represented as an external referee's acceptance. 19
 20

21 WHAT THE REVISION ESTABLISHES 21

22 The revised paper provides a complete, explicitly conditioned construction from 22
 23 trainable neural continuation factors to a full-policy loss target, now including 23
 24 a precision-adaptive refresh rule and an executed exact comparison. It preserves 24
 25 the original economic subject, applications, and unfavorable outcomes. It does not 25
 26 manufacture a broad superiority frontier, an empirical calibration, or a population 26
 27 reliability estimate from the available records. Those distinctions allow the next 27
 28 referee to assess the strengthened mathematical and computational contribution on 28
 29 its actual evidence. 29
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